Hongzheng Tian

Ph.D. Candidate in Computer Engineering, UC Irvine · Graduating June 2027
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Education

University of California, Irvine — Irvine, CA
Ph.D. in Computer Engineering — Sep. 2022 – Jun. 2027 (expected)
M.S. in Electrical and Computer Engineering — Sep. 2020 – Jun. 2022

Northwestern Polytechnical University — Xi'an, China
B.Eng. in Microelectronics Science and Engineering — Sep. 2016 – Jun. 2020

Research & Engineering Experience

UC Irvine — Ph.D. Candidate, CORSA Lab (Advisor: Prof. Sitao Huang) — Sep. 2022 – Present

CoVA: Compiler for Versatile Architectures *— Ph.D. thesis; sole developer; SC24 ACM SRC poster*

- **Compiler:** Built an MLIR-based compiler for NumPy programs: a custom dialect (38 ops) and 30 passes lowering through upstream bufferization to six CPU/GPU routes (LLVM, OpenMP, EmitC, NVVM).
- **Results:** 3.5× geomean CPU speedup over NumPy on 53 NPBench workloads, ahead of Numba and DaCe; 53/54 CPU and 52/54 GPU workloads pass strict golden-reference checks.
- **Analyses & codegen:** Designed region-level storage and alias analyses that prove loop independence and keep buffers GPU-resident; implemented GPU thread-block mapping and deterministic reductions.
- **CUDA runtime:** Implemented stream and scratch-buffer reuse and analysis-guided segmented synchronization that removes unnecessary waits between kernel launches.

HeteroTask: Distributed Heterogeneous Task Runtime *— sole developer; arXiv 2026*

- Built offline DAG scheduling with task-variant profiling, memory-aware device and variant selection, communication-cost modeling, and buffer lifetime planning.
- Implemented a C++/MPI runtime that executes placements and dependency transfers on CPU/GPU nodes.
- Reduced application completion time by 19.7–72.1% relative to StarPU on two 8-GPU nodes.

PyAIE: A Python-based Programming Framework for Versal ACAP *— led design and implementation*

- Generated AI Engine kernels, PL and host code, and configuration scripts for AMD Versal from Python.

Hewlett Packard Enterprise — Research Associate Intern — Milpitas, CA | Jun. – Sep. 2024

HeteroBench: Multi-kernel Benchmarks for Heterogeneous Systems *— lead developer*

- Led a suite of 11 multi-kernel applications (image processing, machine learning, numerical computation, physical simulation) with Python, OpenMP, CUDA, and HLS implementations for CPUs, GPUs, and FPGAs.
- Built a JSON-configured Python driver with Jinja-generated Makefiles for backend-specific builds, execution, profiler integration, and experiment logging.
- ICPE 2025 **Best Paper Award – Artifact Track**; open-sourced as HewlettPackard/HeteroBench.

Hewlett Packard Enterprise — Research Associate Intern — Milpitas, CA | Jun. – Sep. 2023

CoVA: Compiler for Versatile Architectures *— initial framework*

- Built CoVA's initial compilation framework: a custom MLIR dialect separating program semantics from target-specific lowering, MLIR-to-LLVM lowering, and Python bindings.
- Implemented compiler analysis and transformation passes; the project continued as my Ph.D. thesis.

Selected Publications & Patents

- **H. Tian** et al. HeteroBench: Multi-kernel Benchmarks for Heterogeneous Systems. *ICPE 2025*. Best Paper Award (Artifact Track).
- **H. Tian** et al. HeteroTask: Multi-Variant Task Scheduling and Execution on Distributed Heterogeneous Clusters. *arXiv 2026*.
- **H. Tian** et al. FPGA Implementation of Associative Processors. *IEEE TCAS-II*, 2023.
- **H. Tian** et al. PyAIE: A Python-based Programming Framework for Versal ACAP Platforms. *DAC 2023 LBR*.
- A. Mishra, **H. Tian**, et al. Microarchitectural-Neutral Automatic Type and Shape Inference and Cross-Microarchitecture Invocation. *U.S. Patent Application 18/678,875*, allowed.

Technical Skills

Languages: C++, C, Python, Verilog
Compilers: MLIR (dialects, passes, bufferization), LLVM/NVVM, OpenMP offloading, EmitC
Parallel & hardware: OpenMP, MPI, CUDA; NVIDIA GPUs, AMD Versal, FPGAs (Vitis HLS, Vivado)